

FIRST-BAD CUBIC RIGIDITY AND SOURCE-EXPLICIT CROSS-PARITY SECULAR TRANSFER IN THE WEIL-CCM HIERARCHY

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ABSTRACT. A preceding finite-detection theorem shows that every hypothetical nontrivial zero of the Riemann zeta function off the critical line forces a strictly negative direction in an actual canonical centered Weil-CCM matrix, and that this finite badness persists under centered enlargement. We study the first size at which such a failure can occur. At the least globally bad size for a fixed aperture, both reversal-parity predecessor sectors are nonnegative, while one successor parity carries a genuine negative compressed eigenmode. The successor parity space splits as the inherited predecessor space plus a one-dimensional intrinsic shell, and the negative eigenmode has a nonzero component in that shell. A canonical cubic generator meets the same shell nontrivially, while the even-to-odd compression-intertwining defect factors through a single cubic channel.

For every negative spectral parameter, predecessor nonnegativity makes the shifted predecessor block invertible. We use its resolvent to construct a canonical trial vector whose full eigenvalue residual has zero predecessor component. Since the remaining shell is one-dimensional, the residual is exactly a scalar multiple of the canonical cubic shell vector. This gives an exact secular scalar $F_p(\lambda)$ satisfying

$$F_p(\lambda) = 0 \iff \text{the canonical trial vector is a nonzero eigenmode at } \lambda,$$

and hence a genuine two-way scalar characterization of negative compressed eigenvalues. The same scalar is identified with a normalized explicit Schur expression. At zero shift, the projected predecessor block has an exact kernel-range decomposition. The cubic coupling therefore falls into a decoupled branch whose canonical shell-response scalar has strictly negative real part, or a resonant branch with the exact denominator-free pole identity

$$(-\lambda)\Pi_0(R_\lambda b) = \Pi_0(b).$$

Finally, the cubic parity-defect coefficient is identified with an actual quadratic-normal moment of the canonical source matrix, and the even and odd secular scalars satisfy the exact source-explicit transfer law

$$F_-(\lambda) = \alpha(\lambda)F_+(\lambda) + \Gamma(\lambda)\Phi_{N+1,L}(u_+(\lambda)).$$

These are finite counterexample-rigidity theorems. They do not exclude the forced negative root, do not prove finite positivity or a finite-to-infinite positivity closure, and do not prove the Riemann hypothesis. The theorem core is formally verified in Lean 4.

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1. INTRODUCTION

Weil's explicit formula admits a positivity formulation equivalent to the Riemann hypothesis, and finite shadows of that positivity problem have been studied through Hermitian forms, variational principles, structured finite matrices, and operator constructions for decades [15, 16, 1, 2, 3, 5, 4, 13]. Recent finite work also gives an exact Guinand–Weil dictionary on centered Galerkin sectors [6]. Two companion constructions isolate, respectively, a pair-local explicit Gram witness and a complete two-translate detector for hypothetical off-critical data [7, 8].

The present paper begins after the finite-detection problem. In [11], a hypothetical off-critical zero is shown to force a nonzero vector in a canonical centered boundary-flat sector with strictly negative quadratic value. Centered zero extension preserves the represented finite test, all three boundary moments, and the canonical quadratic value, so one bad centered size remains bad at every larger centered size at the same aperture. That result answers the existence question

$$(1) \quad \text{off-critical zero} \implies \text{negative canonical finite direction.}$$

It deliberately leaves open the finer structure of the *first* finite failure. In particular, its open problems ask for a least-bad constrained size, a genuine constrained negative eigenmode, and an exact one-dimensional Schur, Feshbach, KKT, or secular condition for the first instability [11].

The present paper closes that structural problem and then continues beyond it. The first bad state is not merely a finite matrix with a negative Rayleigh direction. After reversal-parity decomposition, global minimality forces both predecessor parity sectors to be nonnegative. The bad successor parity has exactly one new intrinsic degree of freedom over its predecessor, and the negative eigenmode must use it. The same one-dimensional shell is met by a canonical cubic channel already present in the finite source geometry. This reduces the negative-shift spectral problem to one exact scalar residual coordinate.

The resulting chain is

$$(2) \quad \begin{array}{c} \text{hypothetical off-critical zero} \\ \Downarrow \\ \text{globally first bad finite size} \\ \Downarrow \\ \text{negative compressed eigenmode} \\ \Downarrow \\ \text{nonzero one-dimensional cubic shell component} \\ \Downarrow \\ F_p(\lambda) = 0, \quad \lambda < 0. \end{array}$$

The secular scalar is not introduced as an arbitrary inner product. It is the canonical quotient coordinate of the *full residual* after the predecessor equation has been solved exactly. This distinction is essential: vanishing of one scalar inner product does not in general force a vector-valued residual to vanish. Here the residual is already known to lie in a one-dimensional shell, so its scalar coordinate is exact.

The analysis then reaches the zero-shift obstruction. The projected predecessor block has the algebraic and orthogonal decomposition

$$W = \text{Ker } H \oplus \text{Ran } H.$$

If the cubic coupling lies in $\text{Ran } H$, a zero-shift preimage exists and the resulting zero-shift Schur endpoint is canonical; at the forced first-bad negative root its shell response has strictly negative real part. If the cubic coupling has a nonzero kernel component, the shifted resolvent carries an exact pole coefficient rather than merely a qualitative loss of invertibility.

Finally, the parity-defect coefficient is returned to the canonical finite source. The even normal plane is generated by the constant and quadratic centered power vectors. Orthogonalizing the quadratic normal against the constant normal isolates a scalar moment of the canonical source matrix. That moment is exactly the coefficient of the surviving cubic parity channel. At a common safe negative shift the even and odd secular scalars then obey an exact source-explicit transfer law.

Main contributions. The paper establishes the following finite theorem suite.

- (i) A globally first-bad reduction: a hypothetical off-critical zero forces a least bad centered size whose two parity predecessors are both nonnegative.
- (ii) Extraction of a genuine negative compressed eigenmode in one successor parity.
- (iii) An intrinsic predecessor-shell decomposition in which the successor shell has complex dimension one and the first-bad eigenmode has nonzero shell component.
- (iv) A canonical cubic shell direction and an exact one-channel factorization of the even-to-odd compression defect. The resulting defect has rank at most one; exact rank one is not asserted.
- (v) A canonical negative-shift trial vector and an exact secular scalar $F_p(\lambda)$ whose zeros are equivalent to genuine nonzero eigenmodes at $\lambda < 0$.
- (vi) Identification of the quotient secular scalar with a normalized explicit Schur expression, together with shifted coercivity and resolvent metric control.
- (vii) An exact zero-shift kernel-range dichotomy. The decoupled branch has a canonical shell response with strictly negative real part, while the resonant branch has an exact denominator-free kernel-pole identity.
- (viii) Identification of the cubic parity-defect coefficient with a quadratic-normal moment of the actual canonical finite source matrix.
- (ix) An exact source-explicit cross-parity secular transfer

$$F_-(\lambda) = \alpha(\lambda)F_+(\lambda) + \Gamma(\lambda)\Phi(\lambda),$$

with a metric overlap formula for Γ and exact specializations at a forced even or odd secular root.

- (x) Composition of the full finite rigidity package with the preceding off-critical-zero detection theorem.

Claim boundary. Nothing in this paper proves the Riemann hypothesis. We do not prove that either zero-shift branch is impossible; do not prove that α , Γ , the overlap coefficient, or the source moment is nonzero or has a useful sign at the forced state; do not exclude the forced negative secular root; do not establish universal nonnegativity of the canonical finite matrices; and do not prove a finite-to-infinite positivity closure. At every nontrivial size $N \geq 1$, the centered-index parity map used below is an algebraic equivalence on the constrained parity sectors; it is not asserted to be unitary or isometric. Likewise, the cubic defect is proved to factor through a single nonzero generator, but the scalar coefficient functional is not proved nonzero, so rank at most one is not silently upgraded to exact rank one.

2. CANONICAL CENTERED HIERARCHY AND REVERSAL PARITY

We use the canonical finite normalization and boundary-flat coefficient spaces of [10, 11]. The analytic construction of the finite matrix is not repeated here; only the finite properties needed for the first-bad analysis are recalled.

Fix $L > 0$. For $N \in \mathbb{N}$, write

$$A_{N,L} \in M_{2N+1}(\mathbb{C})$$

for the canonical direct-source finite Weil–CCM matrix, indexed by the centered frequencies $-N, \dots, N$. For a coefficient vector $u = (u_{-N}, \dots, u_N)$ define the centered moments

$$(3) \quad M_j^{(N)}(u) := \sum_{n=-N}^N n^j u_n, \quad j \geq 0,$$

and the boundary-flat sector

$$(4) \quad V_N := \{u \in \mathbb{C}^{2N+1} : M_0^{(N)}(u) = M_1^{(N)}(u) = M_2^{(N)}(u) = 0\}.$$

The canonical quadratic form is

$$(5) \quad Q_{N,L}^{\text{can}}(u) := u^* A_{N,L} u.$$

On V_N it agrees exactly with the complete Weil self-form of the represented hard-window finite test [11].

Let J_N be centered reversal,

$$(6) \quad (J_N u)_n := u_{-n}.$$

The canonical matrix commutes with reversal, so V_N decomposes orthogonally into its reversal-parity sectors

$$(7) \quad V_N = V_N^+ \oplus V_N^-, \quad V_N^\pm := \{u \in V_N : J_N u = \pm u\}.$$

Let $P_N^\pm$ denote the orthogonal projections onto these spaces, and define the compressed self-adjoint operators

$$(8) \quad T_{\pm,N,L} := P_N^\pm A_{N,L} |_{V_N^\pm}.$$

For later normal-space calculations, let

$$(9) \quad d_N^j := (n^j)_{n=-N}^N \in \mathbb{C}^{2N+1}, \quad j \geq 0.$$

Within the even ambient parity space, the nonautomatic boundary conditions are the zeroth and second moments, while in the odd ambient parity space the only nonautomatic condition is the first moment. Thus the normal spaces have the concrete forms

$$(10) \quad (V_N^+)^\perp, \text{even} = \text{span}\{d_N^0, d_N^2\}, \quad (V_N^-)^\perp, \text{odd} = \text{span}\{d_N^1\}.$$

In particular, for $N \geq 1$,

$$(11) \quad \dim_{\mathbb{C}} V_N^+ = \dim_{\mathbb{C}} V_N^- = N - 1.$$

This equal-dimension parity geometry is one of the structural inputs already present in the formal finite hierarchy.

Let

$$(12) \quad E_N : \mathbb{C}^{2N+1} \longrightarrow \mathbb{C}^{2(N+1)+1}$$

be centered zero extension. It preserves reversal parity and all centered moments, so

$$(13) \quad E_N(V_N^p) \subseteq V_{N+1}^p, \quad p \in \{+, -\}.$$

For each parity define the predecessor image

$$(14) \quad W_{p,N} := E_N(V_N^p) \subset V_{N+1}^p$$

and its intrinsic orthogonal successor shell

$$(15) \quad S_{p,N} := W_{p,N}^\perp \cap V_{N+1}^p.$$

By (11),

$$(16) \quad \dim_{\mathbb{C}} W_{p,N} = N - 1, \quad \dim_{\mathbb{C}} V_{N+1}^p = N,$$

so $S_{p,N}$ is expected to be one-dimensional. Section 3 records the exact intrinsic statement in the first-bad setting.

Inner-product convention. Throughout the complex finite-dimensional discussion, $\langle x, y \rangle$ is conjugate-linear in the first argument and linear in the second. This is the convention used by the Lean formalization and is relevant to the conjugation in the explicit Schur bridge below.

3. GLOBAL FIRST BADNESS AND CONSTRAINED SPECTRAL EXTRACTION

We first isolate the canonical finite state at which the later scalar structure is forced.

Definition 3.1 (Parity badness). *For $p \in \{+, -\}$, say that (L, N, p) is bad if there exists $0 \neq v \in V_N^p$ with*

$$\operatorname{Re} \langle T_{p,N,L} v, v \rangle < 0.$$

Write $\operatorname{Bad}(L, N)$ if at least one parity is bad.

Because $A_{N,L}$ is reversal symmetric, any negative direction in V_N yields a negative direction in at least one parity sector. The preceding finite-detection theorem therefore implies the following.

Theorem 3.2 (Finite counterexample input). *Assume that the Riemann zeta function has a nontrivial zero ρ_0 with $\operatorname{Re} \rho_0 \neq \frac{1}{2}$. Then there exist $L > 0$ and $N_0 \geq 2$ such that $\operatorname{Bad}(L, N_0)$.*

Proof. By [11], the off-critical zero forces a nonzero $u \in V_{N_0}$ with $Q_{N_0,L}^{\text{can}}(u) < 0$ for some $L > 0$ and $N_0 \geq 2$. Decompose $u = u_+ + u_-$ according to (7). Reversal symmetry kills the mixed quadratic terms, so the negative total value forces a negative value in at least one parity component. $\square$

Definition 3.3 (Globally first bad size). *For a fixed $L > 0$ with at least one bad finite size, let*

$$(17) \quad N_*(L) := \min\{M \geq 0 : \operatorname{Bad}(L, M)\}.$$

The boundary-flat constraints give $V_0 = V_1 = \{0\}$, so no bad size exists below 2. Hence $N_(L) \geq 2$ and we may write*

$$(18) \quad N_*(L) = N + 1, \quad N \geq 1.$$

We call N the predecessor size and $N + 1$ the globally first-bad successor size.

Proposition 3.4 (Both predecessor parities are nonnegative). *At a globally first-bad successor $N + 1$, for each $q \in \{+, -\}$ and every $x \in V_N^q$,*

$$(19) \quad \operatorname{Re} \langle T_{q,N,L} x, x \rangle \geq 0.$$

Moreover, every centered size $M < N + 1$ is globally nonbad.

Proof. This is immediate from the minimality in (17). If either parity at size N admitted a negative direction, then N itself would be bad, contradicting $N < N + 1 = N_*(L)$. $\square$

Choose a parity $p \in \{+, -\}$ for which the successor is bad. Since $T_{p,N+1,L}$ is a self-adjoint operator on the finite-dimensional space V_{N+1}^p , a negative Rayleigh direction yields a negative eigenvalue.

Theorem 3.5 (Negative constrained eigenmode). *There exist $\lambda \in \mathbb{R}$ and $0 \neq v \in V_{N+1}^p$ such that*

$$(20) \quad \lambda < 0, \quad T_{p,N+1,L} v = \lambda v.$$

Proof. The minimum of the Rayleigh quotient of the self-adjoint compression is an eigenvalue. Since the sector is bad, this minimum is strictly negative. $\square$

Proposition 3.6 (Intrinsic predecessor-shell decomposition). *For each parity p and $N \geq 1$,*

$$(21) \quad V_{N+1}^p = W_{p,N} \oplus S_{p,N}, \quad \dim_{\mathbb{C}} S_{p,N} = 1.$$

If v is the negative first-bad eigenmode of Theorem 3.5, then its decomposition

$$(22) \quad v = w + s, \quad w \in W_{p,N}, \quad s \in S_{p,N},$$

has $s \neq 0$.

Proof. The definition (15) gives the orthogonal sum. The dimension count (16) shows that the shell is one-dimensional. If $s = 0$, then $v \in W_{p,N}$ is an embedded predecessor vector. Centered nesting identifies its quadratic value at size $N + 1$ with the predecessor quadratic value at size N , which is nonnegative by Proposition 3.4. On the other hand, (20) gives

$$\operatorname{Re} \langle T_{p,N+1,L} v, v \rangle = \lambda \|v\|^2 < 0,$$

a contradiction. $\square$

The eigenmode also satisfies the exact parity-constrained KKT residual obtained by viewing the compressed eigenvalue equation inside the ambient centered coefficient space. We do not need the explicit residual formula below, but it is part of the formally verified first-bad endpoint and gives the connection to the finite moment-normal geometry used in Sections 4 and 8.

4. THE CUBIC ONE-CHANNEL PARITY DEFECT

The first-bad shell is one-dimensional for purely dimension-theoretic reasons. The additional fact needed for the secular theory is that a canonical cubic vector meets that shell nontrivially and that the even-to-odd finite compression defect is carried by the same cubic geometry.

For a generic finite size $K \geq 2$, let

$$(23) \quad \tilde{D}_K : \mathbb{C}^{2K+1} \longrightarrow \mathbb{C}^{2K+1}, \quad (\tilde{D}_K v)_n := nv_n,$$

be ambient centered-index multiplication. The boundary moments in (4) show that its restriction

$$(24) \quad D_K := \tilde{D}_K|_{V_K^+} : V_K^+ \longrightarrow V_K^-$$

is an algebraic linear equivalence between the equal-dimensional constrained parity sectors. Indeed, if $D_K v = 0$, then $v_n = 0$ for $n \neq 0$, while $M_0^{(K)}(v) = 0$ forces $v_0 = 0$; thus the restriction is injective, and equal dimensions give surjectivity. No metric isometry is asserted.

Let

$$(25) \quad g_K := P_K^- d_K^3 \in V_K^-.$$

The exact odd normal space in (10) is generated by d_K^1 , and for $K \geq 2$ the projected cubic vector is nonzero.

Proposition 4.1 (Cubic generator). *For every $K \geq 2$,*

$$(26) \quad g_K \neq 0.$$

Proof. If $P_K^- d_K^3 = 0$, then d_K^3 would lie in the odd normal line $\text{span}\{d_K^1\}$. Thus $n^3 = cn$ on every centered grid point. At $n = 1$ this gives $c = 1$, while at $n = 2$ it gives $8 = 2$, impossible when $K \geq 2$. $\square$

Define the even-to-odd compressed intertwining defect

$$(27) \quad \mathcal{D}_{K,L} := T_{-,K,L} D_K - D_K T_{+,K,L} : V_K^+ \longrightarrow V_K^-.$$

Since $g_K \neq 0$, define its canonical defect coordinate by

$$(28) \quad \ell_{K,L}(v) := \frac{\langle g_K, \mathcal{D}_{K,L} v \rangle}{\langle g_K, g_K \rangle}, \quad v \in V_K^+.$$

With our inner-product convention this is complex-linear in v .

Lemma 4.2 (Even centered-index commutation). *For $L > 0$, $K \geq 2$, and $v \in V_K^+$,*

$$(29) \quad A_{K,L}(D_K v) = \tilde{D}_K(A_{K,L} v).$$

Proof. This is the even boundary-flat specialization of the exact displacement identity for the canonical source matrix. In raw coefficient form the ambient displacement term $(A_{K,L} \tilde{D}_K - \tilde{D}_K A_{K,L})v$ collapses to the boundary-moment channels; the three conditions $M_0^{(K)}(v) = M_1^{(K)}(v) = M_2^{(K)}(v) = 0$, together with even parity, make that residual vanish identically. The identity is part of the canonical finite source package used by the preceding detection paper and is independently theorem-checked in the Lean source snapshot of Section 12. $\square$

Theorem 4.3 (Exact cubic factorization). *For $L > 0$, $K \geq 2$, and every $v \in V_K^+$, the functional (28) satisfies*

$$(30) \quad \mathcal{D}_{K,L} v = \ell_{K,L}(v) g_K.$$

Consequently

$$(31) \quad \text{rank } \mathcal{D}_{K,L} \leq 1.$$

No nonzeroness of $\ell_{K,L}$ is asserted.

Proof. The source-side reason is the parity-normal geometry. For even $v \in V_K^+$, the difference between the canonical source image $A_{K,L}v$ and its even compression lies in the even normal plane $\text{span}\{d_K^0, d_K^2\}$. Write that residual as

$$(32) \quad A_{K,L}v = T_{+,K,L}v + a_0 d_K^0 + a_2 d_K^2.$$

By Lemma 4.2, applying $\tilde{D}_K$ to (32) gives

$$A_{K,L}(D_K v) = \tilde{D}_K(T_{+,K,L}v) + a_0 d_K^1 + a_2 d_K^3.$$

Since $T_{+,K,L}v \in V_K^+$, the first term on the right is $D_K(T_{+,K,L}v)$. Projecting to V_K^- kills the odd normal vector d_K^1 and sends d_K^3 to g_K . Hence

$$T_{-,K,L}(D_K v) - D_K(T_{+,K,L}v) = a_2 g_K.$$

The preceding equation shows that $\mathcal{D}_{K,L}v = a_2 g_K$. Pairing with g_K and using $g_K \neq 0$ gives $\ell_{K,L}(v) = a_2$ by (28). Section 8 then identifies this canonical coefficient with an explicit source moment. $\square$

We now return to a first-bad successor $K = N + 1$. The odd cubic generator g_{N+1} has a canonical even counterpart obtained by algebraic pullback through D_{N+1} . Denote these two parity generators by

$$(33) \quad g_{-,N} := g_{N+1}, \quad g_{+,N} := D_{N+1}^{-1} g_{N+1}.$$

Their intrinsic shell projections are

$$(34) \quad c_{p,N} := P_{S_{p,N}} g_{p,N} \in S_{p,N}.$$

Theorem 4.4 (Cubic shell incidence). *For $N \geq 1$ and either reversal parity p ,*

$$(35) \quad c_{p,N} \neq 0.$$

Since $S_{p,N}$ is one-dimensional, $c_{p,N}$ is a canonical generator of the entire successor shell.

Proof. Set $K = N + 1$, so $K \geq 2$. Consider first the odd generator $g_K = P_K^- d_K^3$. Every vector in the centered predecessor image $W_{-,N}$ vanishes at the new outer coordinate $n = K$. Suppose, for contradiction, that the shell projection of g_K vanished. Then $g_K \in W_{-,N}$, hence $(g_K)_K = 0$.

Because g_K is the constrained orthogonal projection of d_K^3 , the residual $d_K^3 - g_K$ lies in the exact odd normal line $\text{span}\{d_K^1\}$. Thus $d_K^3 - g_K = a d_K^1$ for some $a \in \mathbb{C}$. Evaluating at the outer coordinate $n = K$ and using $(g_K)_K = 0$ gives $a = K^2$, so

$$g_K = d_K^3 - K^2 d_K^1.$$

But $g_K \in V_K^-$ must have vanishing first centered moment, whereas

$$M_1^{(K)}(g_K) = \sum_{n=-K}^K n^2(n^2 - K^2).$$

Every summand is nonpositive and the $n = 1$ summand is strictly negative for $K \geq 2$. Hence the sum is strictly negative, a contradiction. Therefore the odd cubic generator is not inherited, so its shell projection is nonzero.

For the even generator, suppose $g_{+,N}$ were inherited. Centered-index multiplication commutes with centered zero extension, so the algebraic equivalence D_{N+1} sends the even predecessor image into the odd predecessor image. Since $D_{N+1} g_{+,N} = g_{-,N}$ by definition, this would make the odd generator inherited, contradicting the preceding paragraph. Thus neither parity generator is inherited. In the orthogonal

decomposition $V_{N+1}^p = W_{p,N} \oplus S_{p,N}$, vanishing shell projection is equivalent to membership in $W_{p,N}$, so $c_{p,N} \neq 0$ for both parities. $\square$

5. SHIFTED PREDECESSOR BLOCK AND EXACT SECULAR SCALAR

Fix from now on a first-bad state (L, N, p) , and suppress these indices when no confusion can arise. Put

$$(36) \quad W := W_{p,N}, \quad S := S_{p,N}, \quad c := c_{p,N}.$$

Let

$$(37) \quad T := T_{p,N+1,L} : V_{N+1}^p \longrightarrow V_{N+1}^p.$$

Relative to the orthogonal decomposition $V_{N+1}^p = W \oplus S$, define the projected predecessor block

$$(38) \quad H := P_W T|_W : W \longrightarrow W$$

and the shell-to-predecessor coupling

$$(39) \quad b := P_W T c \in W.$$

We deliberately use H rather than A for the predecessor block, since $A_{N,L}$ already denotes the canonical finite source matrix.

By Proposition 3.4, the predecessor form is nonnegative. The projected block inherits self-adjointness from T .

Lemma 5.1 (Shifted coercivity). *For every $w \in W$ and every $\lambda < 0$,*

$$(40) \quad (-\lambda) \|w\|^2 \leq \operatorname{Re} \langle (H - \lambda I)w, w \rangle.$$

In particular, $H - \lambda I$ is injective, hence invertible in finite dimension.

Proof. Predecessor nonnegativity gives $\operatorname{Re} \langle Hw, w \rangle \geq 0$. Therefore

$$\operatorname{Re} \langle (H - \lambda I)w, w \rangle = \operatorname{Re} \langle Hw, w \rangle - \lambda \|w\|^2 \geq (-\lambda) \|w\|^2.$$

If $(H - \lambda I)w = 0$, the left side of (40) is zero, so $w = 0$ because $-\lambda > 0$. $\square$

For $\lambda < 0$ define the safe shifted predecessor resolvent

$$(41) \quad R_\lambda := (H - \lambda I)^{-1} : W \longrightarrow W.$$

The canonical trial vector is

$$(42) \quad u_p(\lambda) := -R_\lambda b + c.$$

Its shell coordinate is exactly c , so $u_p(\lambda) \neq 0$.

Proposition 5.2 (Exact predecessor cancellation). *For every $\lambda < 0$,*

$$(43) \quad P_W(T - \lambda I)u_p(\lambda) = 0.$$

Proof. Using $P_W(Tc) = b$ and the definition of H ,

$$\begin{aligned} P_W(T - \lambda I)u_p(\lambda) &= -(H - \lambda I)R_\lambda b + b \\ &= -b + b = 0. \end{aligned}$$

$\square$

Thus the full residual lies in the one-dimensional shell S . Since c is a nonzero generator of S , there is a unique scalar $F_p(\lambda)$ such that

$$(44) \quad (T - \lambda I)u_p(\lambda) = F_p(\lambda)c.$$

Equivalently, F_p is the canonical cubic quotient coordinate of the full residual.

Definition 5.3 (Canonical cubic secular scalar). *For $\lambda < 0$, define $F_p(\lambda)$ by (44).*

Theorem 5.4 (Exact secular equivalence). *For every $\lambda < 0$,*

$$(45) \quad F_p(\lambda) = 0 \iff (T - \lambda I)u_p(\lambda) = 0.$$

Hence

$$(46) \quad F_p(\lambda) = 0 \iff u_p(\lambda) \text{ is a nonzero eigenmode of } T \text{ at } \lambda.$$

Moreover,

$$(47) \quad F_p(\lambda) = 0 \iff \exists 0 \neq v \in V_{N+1}^p : Tv = \lambda v.$$

Proof. The first equivalence is immediate from (44) and $c \neq 0$. The trial vector is nonzero because its shell part is c , giving (46).

For the final statement, suppose $0 \neq v$ satisfies $Tv = \lambda v$ with $\lambda < 0$. Its shell component cannot vanish: otherwise $v \in W$, contradicting predecessor nonnegativity as in Proposition 3.6. Since S is one-dimensional, rescale v so that its shell component is exactly c . Writing the normalized vector as $w + c$, the predecessor projection of the eigenvalue equation gives

$$(H - \lambda I)w = -b.$$

Invertibility forces $w = -R_\lambda b$, so the normalized eigenmode is exactly $u_p(\lambda)$. Therefore every genuine negative eigenmode canonically normalizes to the trial vector, proving (47). $\square$

Corollary 5.5 (Forced first-bad root). *At the negative eigenvalue λ of Theorem 3.5,*

$$(48) \quad F_p(\lambda) = 0.$$

6. EXPLICIT SCHUR BRIDGE AND METRIC CONTROL

The quotient definition of F_p is the logically safe scalarization because it is taken after the full residual has been proved to lie in a one-dimensional space. We now identify it with the familiar Schur expression.

For $\lambda < 0$, define

$$(49) \quad \mathcal{S}_p(\lambda) := \langle Tc, c \rangle - \lambda \langle c, c \rangle - \langle R_\lambda b, b \rangle.$$

This orientation is the one used by the formal predecessor-shell calculation. Because our inner product is conjugate-linear in the first slot, the quotient scalar initially matches the conjugate of (49).

Theorem 6.1 (Exact quotient-Schur bridge). *For every $\lambda < 0$,*

$$(50) \quad F_p(\lambda) = \frac{\overline{\mathcal{S}_p(\lambda)}}{\langle c, c \rangle}.$$

Consequently

$$(51) \quad F_p(\lambda) = 0 \iff \mathcal{S}_p(\lambda) = 0.$$

Proof. Take the inner product of the residual in (44) with c in the first slot. Orthogonality of c to W , symmetry of T , and $u_p(\lambda) = -R_\lambda b + c$ give

$$\langle c, (T - \lambda I)u_p(\lambda) \rangle = \overline{\mathcal{S}_p(\lambda)}.$$

On the other hand, (44) and linearity in the second slot give

$$\langle c, (T - \lambda I)u_p(\lambda) \rangle = F_p(\lambda) \langle c, c \rangle.$$

Since $c \neq 0$, its self-inner product is nonzero, yielding (50). $\square$

The projected predecessor block and its negative-shift resolvent are symmetric in the induced ambient inner product. This yields the following metric information.

Proposition 6.2 (Resolvent metric control). *For $\lambda < 0$ and $y \in W$,*

$$(52) \quad 0 \leq \operatorname{Re} \langle R_\lambda y, y \rangle,$$

and

$$(53) \quad (-\lambda) \|R_\lambda y\|^2 \leq \operatorname{Re} \langle R_\lambda y, y \rangle.$$

Moreover $\mathcal{S}_p(\lambda) \in \mathbb{R}$.

Proof. Set $w = R_\lambda y$, so $(H - \lambda I)w = y$. Then (53) is Lemma 5.1 applied to w , and (52) follows because $-\lambda > 0$. Symmetry of $H - \lambda I$ implies symmetry of its inverse. Hence $\langle R_\lambda b, b \rangle$ is real, while the self-adjointness of T makes $\langle Tc, c \rangle$ real and $\langle c, c \rangle$ is real positive. Therefore (49) is real. $\square$

Corollary 6.3 (Unconjugated secular formula). *For $\lambda < 0$,*

$$(54) \quad F_p(\lambda) = \frac{\mathcal{S}_p(\lambda)}{\|c\|^2}.$$

No monotonicity, root uniqueness, or negative-root exclusion is inferred from this representation.

7. ZERO-SHIFT KERNEL–RANGE RIGIDITY

The safe negative-shift resolvent does not extend automatically to $\lambda = 0$. The correct zero-shift analysis is instead controlled by the finite kernel–range geometry of H .

Proposition 7.1 (Kernel–range decomposition). *The symmetric projected predecessor block satisfies*

$$(55) \quad \operatorname{Ran} H \perp \operatorname{Ker} H$$

and

$$(56) \quad W = \operatorname{Ker} H \oplus \operatorname{Ran} H.$$

Proof. If $y = Hx \in \operatorname{Ran} H$ and $z \in \operatorname{Ker} H$, symmetry gives

$$\langle y, z \rangle = \langle Hx, z \rangle = \langle x, Hz \rangle = 0.$$

Thus the range and kernel are orthogonal and have trivial intersection. Finite-dimensional rank–nullity shows that their dimensions sum to $\dim W$, proving (56). $\square$

Let

$$(57) \quad \Pi_0 : W \longrightarrow \text{Ker } H$$

be the projection onto $\text{Ker } H$ along $\text{Ran } H$ associated with (56). The cubic coupling b now has two mutually exclusive possibilities.

7.1. The decoupled branch. Assume

$$(58) \quad \Pi_0(b) = 0.$$

Then $b \in \text{Ran } H$, so there exists $x_0 \in W$ such that

$$(59) \quad Hx_0 = b.$$

If x is another solution, then $x - x_0 \in \text{Ker } H$, and orthogonality to $b \in \text{Ran } H$ gives

$$(60) \quad \langle x, b \rangle = \langle x_0, b \rangle.$$

Thus the scalar

$$(61) \quad \mathcal{S}_{p,0} := \langle Tc, c \rangle - \langle x_0, b \rangle$$

is independent of the selected preimage.

Define the zero-shift trial vector

$$(62) \quad u_{p,0} := -x_0 + c.$$

Proposition 7.2 (Canonical zero-shift endpoint). *In the decoupled branch,*

$$(63) \quad P_W T u_{p,0} = 0$$

and

$$(64) \quad \langle T u_{p,0}, u_{p,0} \rangle = \mathcal{S}_{p,0}.$$

For every $d \in W$,

$$(65) \quad \langle T(u_{p,0} + d), u_{p,0} + d \rangle = \mathcal{S}_{p,0} + \langle Hd, d \rangle.$$

Hence $\text{Re } \mathcal{S}_{p,0}$ is the minimum of the real quadratic form $u \mapsto \text{Re } \langle Tu, u \rangle$ on the affine fiber of vectors with cubic shell coordinate c .

Proof. Equation (63) is the zero-shift version of Proposition 5.2:

$$P_W(Tu_{p,0}) = -Hx_0 + b = 0.$$

The cross terms between $Tu_{p,0}$ and predecessor vectors therefore vanish. Expanding the quadratic form gives (64) and (65). Since H is nonnegative, the final term in (65) is nonnegative. $\square$

Because $Tu_{p,0}$ has zero predecessor component and S is one-dimensional, there is a unique scalar $\sigma_{p,0}$ with

$$(66) \quad Tu_{p,0} = \sigma_{p,0}c.$$

The inner-product convention gives

$$(67) \quad \mathcal{S}_{p,0} = \overline{\sigma_{p,0}} \langle c, c \rangle.$$

Theorem 7.3 (Strict regular response at the forced root). *Suppose the first-bad negative root $\lambda < 0$ lies in the decoupled branch. Then*

$$(68) \quad \operatorname{Re} \mathcal{S}_{p,0} < 0$$

and

$$(69) \quad \operatorname{Re} \sigma_{p,0} < 0.$$

Proof. At the forced root, the normalized negative eigenmode $u_p(\lambda)$ has the same shell coordinate c as $u_{p,0}$. Hence $u_p(\lambda) = u_{p,0} + d$ for some $d \in W$. Its quadratic value is

$$\langle Tu_p(\lambda), u_p(\lambda) \rangle = \lambda \|u_p(\lambda)\|^2 < 0.$$

By (65), this equals

$$\mathcal{S}_{p,0} + \langle Hd, d \rangle$$

with $\operatorname{Re} \langle Hd, d \rangle \geq 0$. Therefore (68) follows. Equation (67) and $\langle c, c \rangle > 0$ give (69). $\square$

This theorem does not make $u_{p,0}$ an eigenvector and does not assert that the whole shell is invariant under T . It concerns only the canonical zero-shift trial vector.

7.2. The resonant branch. Assume instead

$$(70) \quad \Pi_0(b) \neq 0.$$

Then some $z \in \operatorname{Ker} H$ couples nontrivially to b . More importantly, the shifted resolvent carries an exact kernel coefficient.

Theorem 7.4 (Exact denominator-free resonant pole). *For every $\lambda < 0$,*

$$(71) \quad (-\lambda)\Pi_0(R_\lambda b) = \Pi_0(b).$$

Consequently

$$(72) \quad \Pi_0(R_\lambda b) = \frac{1}{-\lambda} \Pi_0(b).$$

In particular, if $\Pi_0(b) \neq 0$, the kernel coordinate of the safe shifted resolvent has an exact $1/(-\lambda)$ pole coefficient.

Proof. The resolvent equation is

$$(H - \lambda I)R_\lambda b = b.$$

Apply the kernel projection Π_0 . Since $\Pi_0(Hy) = 0$ for every $y \in W$, one obtains

$$-\lambda \Pi_0(R_\lambda b) = \Pi_0(b),$$

which is (71). Since $\lambda < 0$, division is legitimate and gives (72). $\square$

The denominator-free identity is the primary theorem. No $\lambda \rightarrow 0^-$ resolvent limit or spectral asymptotic is assumed or proved.

8. SOURCE-EXPLICIT CUBIC DEFECT

The functional $\ell_{K,L}$ in Theorem 4.3 was defined geometrically as the coefficient of the surviving odd cubic channel. We now identify it directly on the canonical finite source side.

Fix $K \geq 2$. Define the centered quadratic normal by orthogonalizing d_K^2 against the constant normal direction:

$$(73) \quad n_{2,K} := d_K^2 - \frac{\langle d_K^0, d_K^2 \rangle}{\langle d_K^0, d_K^0 \rangle} d_K^0.$$

By construction,

$$(74) \quad \langle d_K^0, n_{2,K} \rangle = 0,$$

and $n_{2,K}$ remains in the even normal plane $\text{span}\{d_K^0, d_K^2\}$. For $K \geq 1$ it is nonzero: otherwise d_K^2 would be proportional to the constant vector, contradicted by the centered coordinates 0 and 1.

Definition 8.1 (Quadratic source moment). *For $v \in V_K^+$ define*

$$(75) \quad \Phi_{K,L}(v) := \frac{\langle n_{2,K}, A_{K,L}v \rangle}{\langle n_{2,K}, n_{2,K} \rangle}.$$

Lemma 8.2 (Normal coefficient extraction). *Suppose*

$$(76) \quad A_{K,L}v = T_{+,K,L}v + a_0 d_K^0 + a_2 d_K^2$$

for $v \in V_K^+$. Then

$$(77) \quad \Phi_{K,L}(v) = a_2.$$

Proof. Since $n_{2,K}$ lies in the even normal plane, it is orthogonal to $T_{+,K,L}v \in V_K^+$. Equation (74) kills the a_0 term. Finally, the definition (73) gives

$$\langle n_{2,K}, d_K^2 \rangle = \langle n_{2,K}, n_{2,K} \rangle.$$

Pairing (76) with $n_{2,K}$ in the first slot therefore gives

$$\langle n_{2,K}, A_{K,L}v \rangle = a_2 \langle n_{2,K}, n_{2,K} \rangle,$$

and division by the nonzero denominator proves the claim. $\square$

Theorem 8.3 (Source-explicit cubic defect). *For $L > 0$, $K \geq 2$, and every $v \in V_K^+$,*

$$(78) \quad \boxed{\ell_{K,L}(v) = \Phi_{K,L}(v)}.$$

Equivalently,

$$(79) \quad \mathcal{D}_{K,L}v = \frac{\langle n_{2,K}, A_{K,L}v \rangle}{\langle n_{2,K}, n_{2,K} \rangle} g_K.$$

Proof. Choose the exact normal decomposition (76). The proof of Theorem 4.3 shows that the same coefficient a_2 is the cubic defect coefficient: centered-index multiplication sends the source residual to

$$a_0 d_K^1 + a_2 d_K^3,$$

and the odd compression kills the d_K^1 normal direction while retaining $a_2 g_K$. Thus $\ell_{K,L}(v) = a_2$. Lemma 8.2 gives $a_2 = \Phi_{K,L}(v)$. $\square$

Remark 8.4. *Theorem 8.3 is an identity for the canonical direct-source matrix. It does not assert that $\Phi_{K,L}(v)$ is nonzero or has a useful sign for a particular trial or eigenvector.*

9. CROSS-PARITY SECULAR TRANSFER

We now work at the common successor size $K = N + 1$ and a common negative shift $\lambda < 0$. Assume, as at a globally first-bad state, that both predecessor parity sectors at size N are nonnegative. In this section write $T_{\pm} := T_{\pm, N+1, L}$ for the two successor compressions, and let $Q_{\pm}$ denote their canonical cubic quotient coordinates. Thus both safe resolvents and both canonical trial vectors exist:

$$(80) \quad u_+(\lambda) = -R_{+, \lambda} b_+ + c_+, \quad u_-(\lambda) = -R_{-, \lambda} b_- + c_-.$$

Write

$$(81) \quad F_+(\lambda), \quad F_-(\lambda)$$

for the exact secular scalars of Section 5.

Let $D := D_{N+1} : V_{N+1}^+ \rightarrow V_{N+1}^-$ be the algebraic centered-index map. The canonical cubic quotient coordinate is preserved by this transport. However, Dc_+ is not equal to c_- : there is a non-droppable predecessor correction. Write

$$(82) \quad Dc_+ = d_W + c_-, \quad d_W \in W_{-, N}.$$

Likewise the full odd cubic generator decomposes as

$$(83) \quad g_- = a + c_-, \quad a \in W_{-, N}.$$

The equal shell coefficient in (82) and (83) is a consequence of the canonical normalization of the one-dimensional quotient coordinate.

Set

$$(84) \quad \phi(\lambda) := \ell_{N+1, L}(u_+(\lambda)) = \Phi_{N+1, L}(u_+(\lambda)),$$

where the second equality is Theorem 8.3. Transporting the even residual through D and using the cubic defect factorization gives an odd residual before predecessor correction of the form

$$(85) \quad f(\lambda) + (F_+(\lambda) + \phi(\lambda))c_-,$$

where the predecessor forcing is

$$(86) \quad f(\lambda) := F_+(\lambda)d_W + \phi(\lambda)a \in W_{-, N}.$$

Let

$$(87) \quad \chi_\lambda : W_{-, N} \longrightarrow \mathbb{C}$$

be the odd safe correction functional obtained by applying T_- to the safe predecessor resolvent and reading the result in the odd cubic quotient coordinate:

$$(88) \quad \chi_\lambda(y) := Q_-(T_- R_{-, \lambda} y),$$

where Q_- denotes the canonical quotient coordinate along c_- . The $\lambda R_{-, \lambda} y$ term has zero quotient coordinate because it is an inherited predecessor vector.

The canonical odd trial vector is reconstructed exactly from the transported even trial by

$$(89) \quad u_-(\lambda) = Du_+(\lambda) - R_{-, \lambda} f(\lambda).$$

No metric property of D is used.

Proposition 9.1 (Preliminary transfer identity). *At every common safe negative shift,*

$$(90) \quad F_-(\lambda) = F_+(\lambda) + \phi(\lambda) - \chi_\lambda(f(\lambda)).$$

Proof. The quotient coordinate of the transported even residual is $F_+ + \phi$ by (85). The correction $-R_{-, \lambda}f$ changes the odd residual by

$$-(T_- - \lambda I)R_{-, \lambda}f.$$

Its inherited $\lambda R_{-, \lambda}f$ part has zero shell quotient, while the $T_- R_{-, \lambda}f$ part contributes $\chi_\lambda(f)$. Taking the exact odd residual quotient gives (90). $\square$

By linearity of χ_λ and (86),

$$(91) \quad \chi_\lambda(f) = F_+(\lambda)\chi_\lambda(d_W) + \phi(\lambda)\chi_\lambda(a).$$

Define

$$(92) \quad \alpha(\lambda) := 1 - \chi_\lambda(d_W), \quad \Gamma(\lambda) := 1 - \chi_\lambda(a).$$

Theorem 9.2 (Exact source-explicit cross-parity secular transfer). *For every $\lambda < 0$ at which both predecessor parity sectors are nonnegative,*

$$(93) \quad \boxed{F_-(\lambda) = \alpha(\lambda)F_+(\lambda) + \Gamma(\lambda)\Phi_{N+1,L}(u_+(\lambda)).}$$

Proof. Substitute (91) into (90), collect the coefficients of F_+ and ϕ , and use (84). $\square$

The coefficient Γ has an exact metric interpretation involving the odd trial vector and the full odd cubic generator.

Theorem 9.3 (Overlap formula for Γ). *For $\lambda < 0$,*

$$(94) \quad \boxed{\Gamma(\lambda) = \frac{\langle u_-(\lambda), g_- \rangle}{\langle c_-, c_- \rangle}.}$$

Proof. Write $g_- = a + c_-$ as in (83), and $u_- = -R_{-, \lambda}b_- + c_-$. Orthogonality of predecessor and shell components removes the mixed shell-predecessor terms. Symmetry of the shifted odd predecessor resolvent identifies the remaining correction with $\chi_\lambda(a)$. The numerator therefore equals

$$\langle c_-, c_- \rangle - \chi_\lambda(a) \langle c_-, c_- \rangle,$$

which is $\Gamma(\lambda) \langle c_-, c_- \rangle$ by (92). Division by the nonzero shell norm gives (94). $\square$

Corollary 9.4 (Even-root specialization). *If $F_+(\lambda) = 0$, then*

$$(95) \quad \boxed{F_-(\lambda) = \Gamma(\lambda)\Phi_{N+1,L}(u_+(\lambda)).}$$

Equivalently,

$$(96) \quad F_-(\lambda) = \frac{\langle u_-(\lambda), g_- \rangle}{\langle c_-, c_- \rangle} \Phi_{N+1,L}(u_+(\lambda)).$$

Corollary 9.5 (Odd-root identity). *If $F_-(\lambda) = 0$, then*

$$(97) \quad \boxed{\alpha(\lambda)F_+(\lambda) + \Gamma(\lambda)\Phi_{N+1,L}(u_+(\lambda)) = 0.}$$

No division by Γ , α , or the source moment is justified without additional information.

10. COMPOSITION WITH A HYPOTHETICAL OFF-CRITICAL ZERO

We can now package the post-detection finite theorem suite at one common globally first-bad state.

Theorem 10.1 (First-bad cubic rigidity from an off-critical zero). *Assume that there exists a nontrivial zero ρ_0 of ζ with $\operatorname{Re} \rho_0 \neq \frac{1}{2}$. Then there exist*

$$(98) \quad L > 0, \quad N \geq 1, \quad p \in \{+, -\}, \quad \lambda < 0,$$

such that all of the following hold at the same finite state.

(a) $N + 1$ is globally first bad:

$$\operatorname{Bad}(L, N + 1), \quad M < N + 1 \implies \neg \operatorname{Bad}(L, M).$$

(b) Both parity predecessors at size N are nonnegative.

(c) The bad successor parity has a nonzero eigenmode at λ and the exact secular root

$$F_p(\lambda) = 0 \iff \mathcal{S}_p(\lambda) = 0.$$

(d) Its negative eigenmode has a nonzero component in the one-dimensional intrinsic cubic shell.

(e) The zero-shift geometry satisfies exactly one of the following two branches:

(i) decoupled: $\Pi_0(b_p) = 0$, a zero-shift preimage exists, and the canonical shell response obeys

$$\operatorname{Re} \sigma_{p,0} < 0;$$

(ii) resonant: $\Pi_0(b_p) \neq 0$ and

$$(-\lambda)\Pi_0(R_{p,\lambda}b_p) = \Pi_0(b_p).$$

(f) At the same L, N, λ , the exact source-explicit cross-parity transfer (93) holds.

(g) If the forced root has even parity, then $F_+(\lambda) = 0$ and the specialization (95) holds. If the forced root has odd parity, then $F_-(\lambda) = 0$ and the undivided zero identity (97) holds.

Proof. Theorem 3.2 and minimality give a globally first-bad successor and Proposition 3.4. Theorem 3.5 gives the negative compressed eigenmode, and Proposition 3.6 together with Theorem 4.4 gives the one-dimensional cubic shell geometry. The exact negative-shift root follows from Theorem 5.4 and Theorem 6.1. Proposition 7.1, Theorem 7.3, and Theorem 7.4 give the exhaustive zero-shift branch classification. Finally, both predecessors are nonnegative at the same state, so Theorem 9.2 applies at the same λ . Corollaries 9.4 and 9.5 give the parity-specific root statements. $\square$

Remark 10.2 (What the theorem does not say). *Theorem 10.1 classifies and transfers the finite obstruction; it does not contradict it. In particular, the theorem does not show that Γ or Φ is nonzero at an even root, does not show that the odd scalar must have a forbidden sign, and does not exclude the resonant branch. The remaining problem is to rule out the forced negative-root configuration by an additional theorem not supplied here.*

11. RELATION TO EXISTING FINITE WEIL CONSTRUCTIONS

The ingredients used above have substantial classical and modern precedent, while the exact composition is narrower.

11.1. Bombieri’s finite negative-inertia theory. Bombieri’s finite spectral model already establishes negative-index phenomena associated with nonreal transformed zero pairs and, under a finiteness hypothesis, eventual finite negative-inertia counts for zeta truncations [1, 2]. The implication “off-critical spectral data produces finite negative geometry” is therefore established territory. The present result addresses a different question: after a counterexample has been transported into the canonical centered Weil–CCM hierarchy, what exact structure is forced at the globally first bad member of that hierarchy?

11.2. Connes–van Suijlekom and the CCM finite program. Connes and van Suijlekom study structured finite quadratic forms and their real-zero consequences [5], while Connes, Consani, and Moscovici develop finite spectral-triple and rank-one-perturbation constructions around the zeta spectrum [4]. These works provide close finite and operator-theoretic context for parity, compression, and low-rank spectral structure. The present paper does not claim novelty for self-adjoint compression, rank-one or rank-at-most-one perturbation mechanisms, resolvents, or scalar secular equations in isolation.

11.3. The exact finite dictionary. Groskin proves an exact finite Guinand–Weil dictionary for real-even centered Galerkin vectors and gives quantitative archimedean tail certification [6]. The earlier manuscripts [9, 10, 11] reconcile the canonical source normalization, close the scalar finite-dictionary defect, and prove counterexample detection and centered persistence. The present paper begins after those representation and detection steps.

11.4. Suzuki and continuous operator formulations. Suzuki develops a unified continuous framework for Weil’s quadratic form through the screw function and nonlocal self-adjoint operators [13]. That work concerns a continuous/operator limit architecture. The present theorem is finite and local in N : it identifies the internal structure of the first bad canonical finite state and makes no finite-to-infinite spectral claim.

11.5. Recent fixed-window parity results. Zhu gives a certified finite reduction for compact-window Weil positivity and proves fixed-scale parity information, including a simple even ground state at the certified window [14]. This is close context for the role of parity, but it concerns a different window-reduction problem and does not state the globally first-bad centered predecessor–shell, exact cubic secular, zero-shift branch, or source-explicit cross-parity transfer results proved here.

12. FORMAL VERIFICATION IN LEAN 4

The theorem core is formalized in the public repository

`HMarcusWH/zeta-23-lean`.

For broader background on formalizing zeta and L -functions in Lean, see [12]. This draft is pinned to the merged default-branch commit

(99) `e1192857afed9f68fa4a13143ce690b62191b997`,

with Git tree

(100) `2f042a3b0b3313e7c67d627a58a32d579d4e7ff7`.

TABLE 1. Comparison with the closest finite frameworks.

Feature	Bombieri 2000/2003	Groskin 2026	Present result
Finite negative geometry from off-critical spectral data	yes	not the stated endpoint	inherited input
Exact canonical centered finite dictionary	different finite object	yes, real-even Galerkin sector	inherited/used
Globally first-bad size with both parity predecessors nonnegative	not this formulation	not this formulation	yes
One-dimensional intrinsic successor shell at first badness	not this formulation	not the stated endpoint	yes
Exact cubic one-channel compression defect	not this formulation	not the stated endpoint	yes
Scalar root iff genuine negative compressed eigenmode	not this formulation	not the stated endpoint	yes
Zero-shift decoupled/resonant rigidity with exact pole coefficient	not this formulation	not the stated endpoint	yes
Source-explicit cross-parity secular transfer	not this formulation	not the stated endpoint	yes
Lean formalization of theorem core	no	no	yes

This is the merge commit of PR #129 on `main`. Its Git tree is identical to the validated theorem head `440be3e5b6bf05e94ae2c65b1704d52d20acc9af`. The RHRC research/Lean suite and the independent Permansson v0.1.6 formal-verification workflow both completed successfully on that theorem head, so the merged snapshot carries the identical proof-bearing tree.

The toolchain is

`leanprover/lean4:v4.33.0-rc2`,

and Mathlib is pinned to

`51e6992efd06126df61a496bebf8f49482a4e129`.

Table 2 records the main paper-to-Lean correspondence. Several paper statements are direct compositions of smaller declarations rather than separately named compiler objects. This is stated explicitly rather than treating a paper-level packaging theorem as if it were an additional formal declaration.

Formal-status boundary. The source-explicit cross-parity wrapper and the regular/resonant branch-response wrapper are separately compiler-verified on the same proof-bearing snapshot. Theorem 10.1 packages both outputs at one common state as a paper-level composition; at this snapshot we do not claim a separately named Lean declaration with exactly the prose conjunction of Theorem 10.1. The substantive source modules contain explicit `#print axioms` audits. The promoted theorem endpoints report only Lean’s standard logical axioms such as `propext`,

TABLE 2. Selected paper results and Lean declarations/modules at the pinned snapshot.

Paper result	Lean declaration or source module
Global first-bad state, both predecessor parities, negative eigenmode, intrinsic shell, cubic factorization	<code>Zeta23.ExceptionalZero.exists_globalFirstBad_intrinsicShell_cubicFactorization_of_offLine_zero</code>
Intrinsic one-dimensional successor shell	<code>Zeta23.CCM.finrank_intrinsicParitySuccShell</code>
Canonical cubic shell nonzeroness	<code>Zeta23.CCM.intrinsicCubicShellPart_ne_zero</code>
Even centered-index source commutation	<code>Zeta23.CCM.canonicalSourceMatrix_toEuclideanLin_commutes_index_on_even</code>
Canonical trial vector and residual	<code>Zeta23.CCM.cubicSecularTrialVector,</code> <code>Zeta23.CCM.cubicSecularResidual</code>
Residual has zero predecessor coordinate	<code>Zeta23.CCM.intrinsicPredecessorPart_cubicSecularResidual_eq_zero</code>
Exact secular scalarization	<code>Zeta23.CCM.cubicSecularScalar_eq_zero_iff_residual_eq_zero</code>
Secular root iff canonical trial eigenmode	<code>Zeta23.CCM.cubicSecularScalar_eq_zero_iff_trial_eigenmode</code>
Secular root iff existence of a nonzero eigenmode	<code>Zeta23.CCM.cubicSecularScalar_eq_zero_iff_exists_eigenmode</code>
Explicit Schur bridge	<code>Zeta23.CCM.cubicSecularScalar_eq_star_cubicExplicitSchurScalar_div</code>
Projected predecessor symmetry and shifted coercivity	<code>Zeta23.CCM.intrinsicPredecessorBlock_isSymmetric,</code> <code>Zeta23.CCM.re_inner_shiftedIntrinsicPredecessorBlock_ge</code>
Kernel-range zero-shift decomposition	<code>Zeta23.CCM.intrinsicPredecessorBlock_kernel_isCompl_range</code>
Canonical zero-shift Schur endpoint	<code>Zeta23.CCM.cubicZeroShiftSchurEndpoint</code>
Pure-shell zero-shift response	<code>Zeta23.CCM.cubicZeroShiftShellResponseScalar_smul_cubic_eq</code>
Exact resonant kernel pole	<code>Zeta23.CCM.neg_lam_smul_intrinsicPredecessorKernelPart_resolvent_eq</code>
Global regular/resonant first-bad wrapper	<code>Zeta23.ExceptionalZero.exists_globalFirstBad_zeroShiftBranchResponseDichotomy_of_offLine_zero</code>
Source-explicit cubic defect	<code>Zeta23.CCM.cubicDefectFunctional_eq_evenQuadraticSourceMoment</code>
Exact source-explicit cross-parity transfer	<code>Zeta23.CCM.cubicSecularScalar_crossParity_source_transfer</code>
Overlap formula for Γ	<code>Zeta23.CCM.crossParitySecularGamma_eq_trial_cubic_overlap_div</code>
Global first-bad cross-parity wrapper	<code>Zeta23.ExceptionalZero.exists_globalFirstBad_crossParitySecularTransfer_of_offLine_zero</code>

`Classical.choice`, and `Quot.sound`; these are not mathematical assumptions about zeta zeros or the Riemann hypothesis.

13. PRIORITY BOUNDARY AND NOVELTY STATEMENT

The broad ingredients are classical. Weil positivity is equivalent to the critical-line statement; Bombieri already establishes finite negative-index phenomena associated with off-critical spectral data; finite-dimensional spectral compression, Schur complements, resolvents, kernel-range decompositions, and scalar secular equations are

standard linear algebra and operator theory; and recent finite Weil work already provides exact dictionaries and structured finite matrix models [1, 2, 5, 4, 6, 13, 14]. No novelty claim is made for those ingredients in isolation.

The narrower claim concerns their exact realization and composition in the canonical centered Weil–CCM hierarchy at a globally first-bad state. In a targeted audit of the closest publicly available materials used in this project, we did not identify a prior theorem combining all of the following:

- (a) one arbitrary hypothetical off-critical zeta zero forcing a globally first-bad canonical centered size with both predecessor parities nonnegative;
- (b) a genuine negative compressed eigenmode whose new component lies in a one-dimensional intrinsic successor shell generated canonically by a cubic direction;
- (c) an exact two-way secular scalar whose zeros are equivalent to genuine negative compressed eigenmodes at safe negative shifts;
- (d) a zero-shift decoupled/resonant rigidity theorem with a regular shell-response scalar of strictly negative real part and an exact denominator-free resonant pole coefficient;
- (e) an identification of the cubic parity-defect functional with an actual quadratic-normal moment of the canonical finite source matrix; and
- (f) an exact source-explicit even-to-odd secular transfer of the form

$$F_- = \alpha F_+ + \Gamma \Phi.$$

We therefore describe the result conservatively as an apparently new first-bad finite rigidity and source-explicit parity-transfer theorem for the canonical centered Weil–CCM hierarchy. This is a “to our knowledge” statement, not a claim of exhaustive universal priority. The individual linear-algebra mechanisms remain classical, and no priority claim is made beyond the exact finite composition stated above.

The directionality of the final transfer is part of the claim boundary. The theorem proved here is an exact even-to-odd transfer at a common negative shift. No symmetric odd-to-even analogue is asserted without a separate proof.

14. OPEN PROBLEMS

The first-bad obstruction has now been reduced to a small number of exact scalar and branch questions.

P1. Nonvanishing of the source channel. At a forced first-bad state, determine whether the even source moment

$$(101) \quad \Phi_{N+1,L}(u_+(\lambda))$$

must be nonzero in the configurations relevant to Corollary 9.4.

P2. Cross-parity coefficient control. Determine whether $\Gamma(\lambda)$, equivalently the overlap

$$\frac{\langle u_-(\lambda), g_- \rangle}{\langle c_-, c_- \rangle},$$

has a forced sign or nonvanishing property at a globally first-bad state. The same question applies to $\alpha(\lambda)$.

P3. Branch exclusion. Determine whether either zero-shift branch in Theorem 10.1 can be ruled out by the canonical source geometry. The decoupled branch already carries a shell-response scalar with strictly negative real part; the resonant branch carries an exact pole coefficient. Neither fact is presently a contradiction.

- P4. Negative-root exclusion. Use the exact secular and cross-parity identities to rule out the forced $\lambda < 0$ root without assuming a finite positivity statement equivalent to the desired conclusion. Root uniqueness would still be weaker than root absence.
- P5. Effective first-bad localization. The detection theorem of [11] is existential in finite size. Combine the present rigidity theory with an effective upper bound for the first detected or first bad size in terms of aperture, negative margin, and detector data.
- P6. General L -functions. Determine which other explicit-formula finite hierarchies admit the same ingredients: a finite counterexample detector, centered nesting, reversal-like parity, a one-dimensional successor shell, a canonical low-degree defect channel, and an exact source-explicit secular transfer.
- P7. Formal consolidation. Add a single theorem-level Lean wrapper packaging the zero-shift branch response and the source-explicit cross-parity certificate at one common off-critical first-bad state, matching Theorem 10.1 verbatim at the compiler level.

15. CONCLUSION

The preceding finite-detection theorem answers where a hypothetical failure of the critical-line statement must enter the canonical centered hierarchy: it must produce an actual negative canonical finite direction, and centered enlargement cannot hide it again. The present paper asks what the *first* such finite failure must look like.

Global first badness makes both predecessor parity sectors nonnegative. The bad successor parity therefore carries a genuine negative eigenmode with a nonzero component in a one-dimensional new shell. The canonical cubic generator meets that shell, and the parity-compression defect factors through the corresponding cubic channel. Solving the predecessor equation at every negative shift leaves a residual entirely in that one-dimensional shell, giving an exact secular scalar

$$F_p(\lambda) = 0 \iff \text{nonzero compressed eigenmode at } \lambda < 0.$$

The scalar is exactly the normalized explicit Schur expression, not merely a necessary inner-product condition.

At zero shift, the projected predecessor block splits into kernel and range. The regular branch has a canonical zero-shift solution and a shell-response scalar with strictly negative real part at the forced root; the resonant branch has an exact kernel pole

$$(-\lambda)\Pi_0(R_\lambda b) = \Pi_0(b).$$

The cubic parity coefficient is also source-explicit: it is the quadratic-normal moment of the actual canonical finite source matrix. Finally, the two parity secular scalars satisfy

$$F_-(\lambda) = \alpha(\lambda)F_+(\lambda) + \Gamma(\lambda)\Phi_{N+1,L}(u_+(\lambda)).$$

Thus the unresolved problem has moved again. It is no longer to show that a hypothetical off-critical zero enters the finite hierarchy, nor merely to produce a scalar reduction of the first instability. The remaining task is to exclude the exact negative-root configuration forced by the finite rigidity theorem above. That exclusion is not established here, and the Riemann hypothesis remains open.

APPENDIX A. COORDINATE FORM OF THE PARITY-NORMAL CALCULATION

For completeness, we record the elementary coordinate calculation behind the one-channel source defect. Let $K \geq 2$ and $v \in V_K^+$. Because the even constrained normal space is $\text{span}\{d_K^0, d_K^2\}$, there exist coefficients a_0, a_2 such that

$$(102) \quad A_{K,L}v = T_{+,K,L}v + a_0 d_K^0 + a_2 d_K^2.$$

Ambient centered-index multiplication satisfies

$$\tilde{D}_K d_K^0 = d_K^1, \quad \tilde{D}_K d_K^2 = d_K^3.$$

On the even constrained sector the canonical source and the centered-index map obey the exact commutation identity used in the formal finite source construction. Hence

$$A_{K,L}(D_K v) = \tilde{D}_K(T_{+,K,L}v) + a_0 d_K^1 + a_2 d_K^3 = D_K(T_{+,K,L}v) + a_0 d_K^1 + a_2 d_K^3.$$

Projecting to V_K^- gives

$$\begin{aligned} T_{-,K,L}(D_K v) &= D_K(T_{+,K,L}v) + a_2 P_K^- d_K^3 \\ &= D_K(T_{+,K,L}v) + a_2 g_K. \end{aligned}$$

This is (30) with $\ell_{K,L}(v) = a_2$.

Now $n_{2,K}$ from (73) is orthogonal to both V_K^+ and d_K^0 , while

$$\langle n_{2,K}, d_K^2 \rangle = \langle n_{2,K}, n_{2,K} \rangle.$$

Pairing (102) with $n_{2,K}$ therefore yields

$$a_2 = \frac{\langle n_{2,K}, A_{K,L}v \rangle}{\langle n_{2,K}, n_{2,K} \rangle} = \Phi_{K,L}(v),$$

which is Theorem 8.3.

APPENDIX B. CROSS-PARITY TRANSFER ALGEBRA

The transfer formula of Section 9 can be summarized in four exact identities. At a common safe negative shift, write

$$\phi := \Phi_{N+1,L}(u_+), \quad F_{\pm} := F_{\pm}(\lambda).$$

Then

$$(103) \quad Dc_+ = d_W + c_-,$$

$$(104) \quad g_- = a + c_-,$$

$$(105) \quad f = F_+ d_W + \phi a,$$

$$(106) \quad u_- = Du_+ - R_{-, \lambda} f.$$

The quotient of the transported even residual before the predecessor correction is $F_+ + \phi$. The correction changes that quotient by $\chi_{\lambda}(f)$. Hence

$$\begin{aligned} F_- &= F_+ + \phi - \chi_{\lambda}(f) \\ &= F_+ + \phi - F_+ \chi_{\lambda}(d_W) - \phi \chi_{\lambda}(a) \\ &= (1 - \chi_{\lambda}(d_W))F_+ + (1 - \chi_{\lambda}(a))\phi. \end{aligned}$$

Defining $\alpha = 1 - \chi_{\lambda}(d_W)$ and $\Gamma = 1 - \chi_{\lambda}(a)$ gives (93). The overlap identity (94) follows by expanding $\langle u_-, g_- \rangle$ under (104) and using predecessor-shell orthogonality and symmetry of the safe odd predecessor resolvent.

APPENDIX C. REPRODUCIBILITY AND DISCLOSURES

Repository. <https://github.com/HMarcusWH/zeta-23-lean>

Proof-bearing source snapshot. `e1192857afed9f68fa4a13143ce690b62191b997`

Git tree. `2f042a3b0b3313e7c67d627a58a32d579d4e7ff7`

Source-state note. The pinned snapshot is the merged PR #129 commit on `main`.

Its tree is identical to the validated theorem head

`440be3e5b6bf05e94ae2c65b1704d52d20acc9af`.

Lean. `leanprover/lean4:v4.33.0-rc2`

Mathlib. `51e6992efd06126df61a496bebf8f49482a4e129`

Selected formal sources.

- `Zeta23/ExceptionalZero/GlobalFirstBadRigidity.lean`
- `Zeta23/CCM/ParityCompressionRankOne.lean`
- `Zeta23/CCM/ParityCubicFactorization.lean`
- `Zeta23/CCM/CubicShellIncidence.lean`
- `Zeta23/CCM/CubicSecularEquation.lean`
- `Zeta23/CCM/CubicExplicitSecular.lean`
- `Zeta23/CCM/CubicSecularMetric.lean`
- `Zeta23/CCM/ZeroShiftSchurDichotomy.lean`
- `Zeta23/CCM/ZeroShiftSchurEndpoint.lean`
- `Zeta23/CCM/ZeroShiftShellResponse.lean`
- `Zeta23/CCM/ZeroShiftBranchResponse.lean`
- `Zeta23/CCM/SourceExplicitCubicDefect.lean`
- `Zeta23/CCM/CrossParitySecularTransferCore.lean`
- `Zeta23/CCM/CrossParitySecularTransfer.lean`
- `Zeta23/ExceptionalZero/GlobalFirstBadCrossParitySecularTransfer.lean`

Validation. On the proof-bearing PR #129 head with the identical Git tree, the RHRC research and Lean checks completed successfully, and the independent Hermansson v0.1.6 formal-verification workflow completed successfully.

Data and code availability. No empirical dataset is required for the theorem. The Lean 4 formalization, finite normalization audits, theorem-control material, and validation scripts are publicly available in the repository at the pinned source snapshot.

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